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PHYSICS

Physics: Principles and Applications — Part 3

Class 12

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Chapter 1

Units and Measurement

Measurement is the comparison of an unknown quantity with a standard. The International System of Units (SI) defines seven base units: metre, kilogram, second, ampere, kelvin, mole and candela. Derived units combine these to express quantities such as force, energy and pressure. Accuracy and precision characterize the quality of measurements, while significant figures communicate their reliability. Dimensional analysis checks the consistency of equations and helps derive relationships between physical quantities.

Chapter 2

Laws of Motion

Newton's laws of motion describe the relationship between a body and the forces acting on it. The first law, the law of inertia, states that an object remains at rest or in uniform motion unless acted upon by a net external force. The second law quantifies this: force equals mass times acceleration. The third law states that every action has an equal and opposite reaction. These laws explain phenomena from projectile motion to rocket propulsion and form the bedrock of classical mechanics.

Chapter 3

Work, Energy and Power

Work is done when a force causes displacement. Energy is the capacity to do work and exists in many forms — kinetic, potential, thermal, chemical and nuclear. The work-energy theorem connects work to changes in kinetic energy. The law of conservation of energy states that energy can neither be created nor destroyed, only transformed. Power is the rate of doing work. These concepts are central to understanding machines, engines and the natural world.